Last Name: ________________ **First Name:** _______________ **Thermo No.** __________

ME 200 Thermodynamics 1
Fall 2019 – Exam 1

Circle your instructor's last name	
Division 1 (7:30): Naik	**Division 2 (10:30): Braun**
Division 3 (1:30): Holloway	**Division 4 (8:30): Choi**
Division 6 (11:30): Buckius	**Division 7 (2:30): Goldenstein**
Division 8 (12:30): Buckius	

Number of <u>extra</u> papers used if any

DISCLAIMER

This practice exam is only for practice. The actual exam may not have a format similar to the problems included in this practice exam.

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1. [16 points] Fill in the circle for the correct answer(s) for each question. There can be more than one correct answer for a question. No justifications are needed on any of these questions.

(a) Internal energy is only a function of temperature for which of the following?

○ saturated liquid vapor mixture ○ compressed liquid ○ incompressible substance ○ superheated vapor ○ ideal gas

(b) Enthalpy is only a function of temperature for which of the following?

○ saturated liquid vapor mixture ○ compressed liquid ○ incompressible substance ○ superheated vapor ○ ideal gas

(c) Which combination of assumptions below are always required to evaluate change in specific internal energy as $u_2 - u_1 = c_v(T_2 - T_1)$?

○ closed system ○ constant specific heats ○ constant volume process ○ constant pressure process ○ ideal gas

(d) A system undergoes a process for which the first law simplifies to:

$$U_2 - U_1 = Q_{12}$$

Which of the following assumptions must be true for this system and process?

○ closed system
○ polytropic process with $n = 0$
○ constant volume process
○ open system
○ adiabatic system
○ negligible ΔKE and ΔPE

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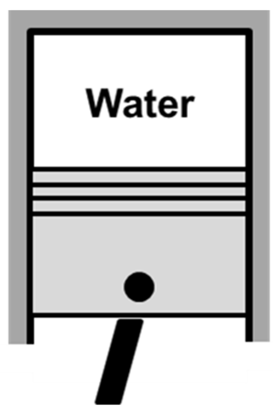
2. [42 points] Water is contained within a piston-cylinder system. The initial temperature and volume are T_1 = 160°C and V_1 = 2.00 m^3, and there are equal volumes of liquid and vapor in this initial state. There is heat transfer at constant temperature until the piston is at state 2 where the pressure is p_2 = 3.00 bar. The work done by the water from state 1 to state 2 is 300 kJ/kg.

(a) Find the initial quality, x_1. Report your answer in %.
(b) Show the process on a P-v diagram relative to the vapor dome and the lines of constant temperature for the two states. Label the two states and indicate the process direction with arrows. For water: $T_{critical}$ = 374°C, $P_{critical}$ = 221 bar.
(c) Calculate the heat transfer per unit mass of water for the process from state 1 to state 2. Report your answer in kJ/kg.

<u>Identify the system on the sketch provided, show mass/energy interactions with directions, list any assumptions and basic equations, and provide your solution. Write your final answers in the boxes below. There is no need to re-write the given and find.</u>



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Extra Space for Problem 2

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Extra Space for Problem 2

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3. [42 points] Air in a piston-cylinder assembly undergoes two processes.

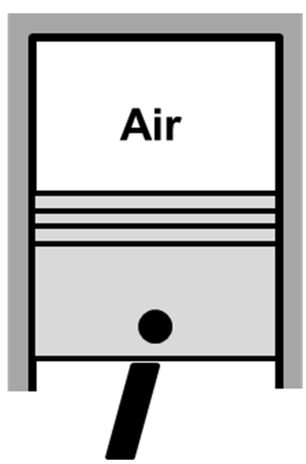
Process 1 to 2: A polytropic process with $n = -2$, Pv^{-2} = constant.
Process 2 to 3: A constant volume process to state 3.

Some state properties for air are provided in the table below.
MW_{air} = 28.97 kg/kmol.

State	***P*** **[bar]**	***T*** **[K]**	***v*** $[m^3/kg]$
1	1.00	300	
2			
3	2.00		1.10

(a) Determine the values of the temperatures at state 2 and 3, T_2 and T_3. Report your answers in K.
(b) Calculate the work per unit mass during the process from state 1 to state 2 and from state 2 to state 3. Report your answers in kJ/kg.
(c) Show the two processes on the *T*-*v* diagram indicating the values of temperature and specific volume.

<u>Identify the system on the sketch provided, show mass/energy interactions with directions, list any assumptions and basic equations, and provide your solution. Write your final answers in the boxes below. There is no need to re-write the given and find.</u>



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Extra Space for Problem 3

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Extra Space for Problem 3

Selected Answers

1(a) **incompressible substance, ideal gas**

1(b) **ideal gas**

1(c) **constant specific heat, ideal gas**

1(d) **closed system, constant volume process, negligible ΔKE and ΔPE**

2(a) $x_1 = 0.358\%$

2(c) $q_{12} = +2205.8$ **kJ/kg**

3(a) $T_2 = 625.5$ **K**, $T_3 = 766.5$ **K**

3(b) $w_{12} = +31.1$ **kJ/kg**, $w_{23} = 0$